

Biophysics (Bio215)

TA:

Bhumika Choudhary (bhumikac@iiitd.ac.in)

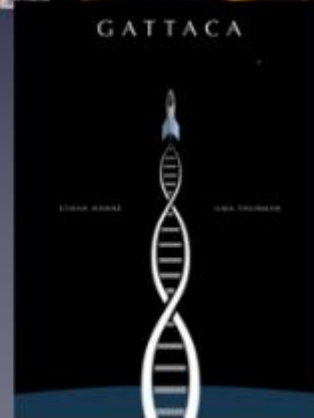
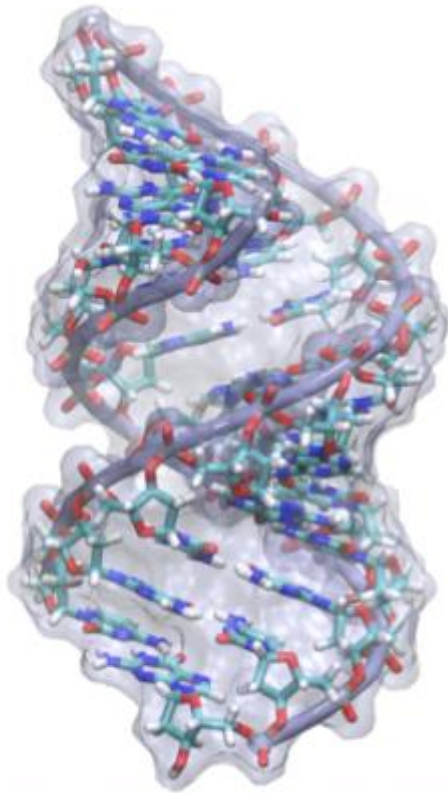
Deeksha Yadav (deekshay@iiitd.ac.in)

Dr. Arjun Ray
(arjun@iiitd.ac.in)

Today's lecture

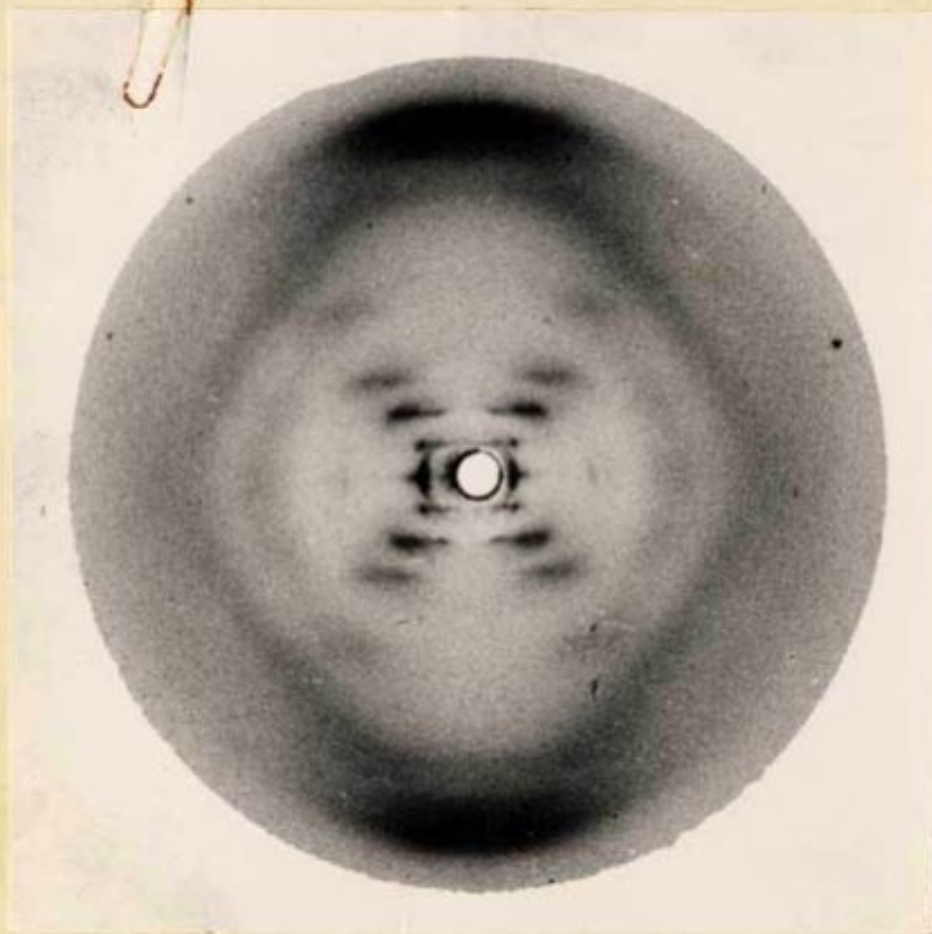
- DNA & RNA
- central dogma of molecular biology
- X-ray crystallography
- Cryo-electron microscopy (cryo-EM)
- Protein structure & function
- Water & Hydrogen bonds

DNA - the molecule of life





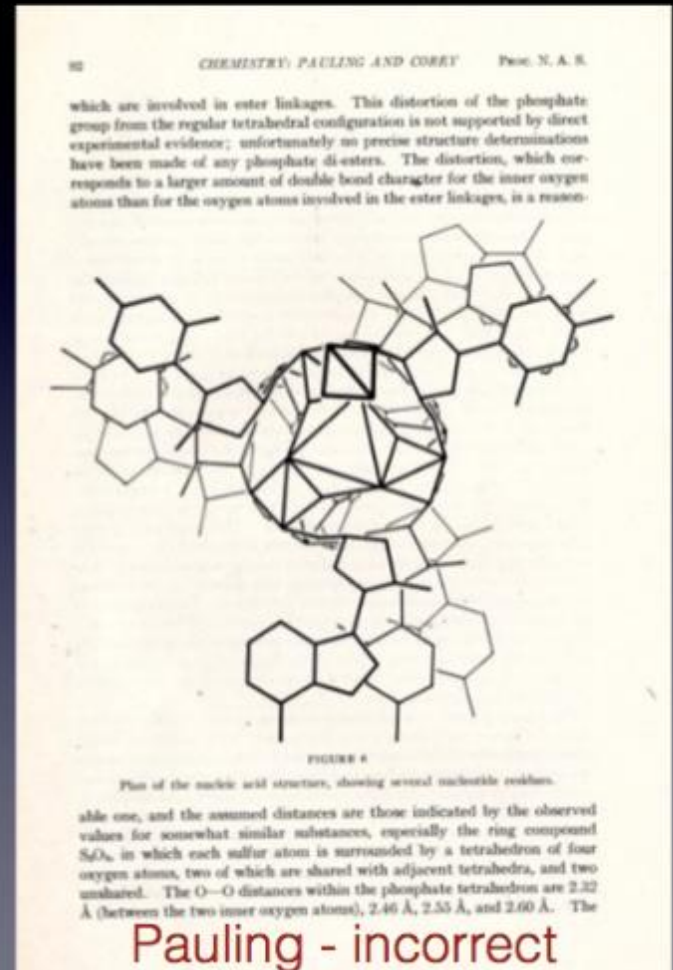
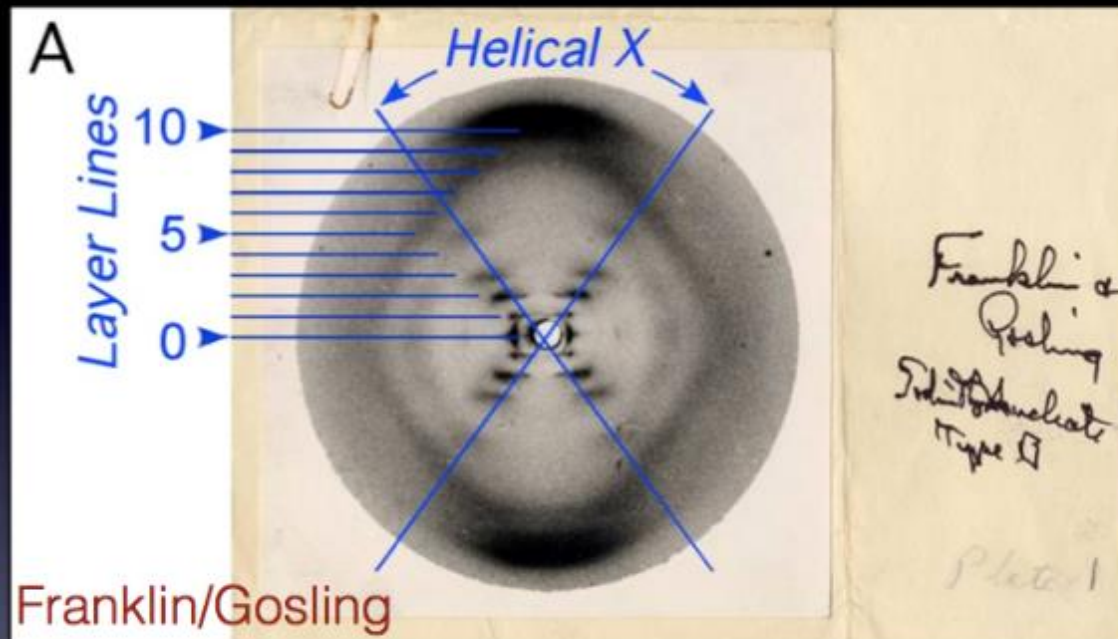
Rosalind Franklin

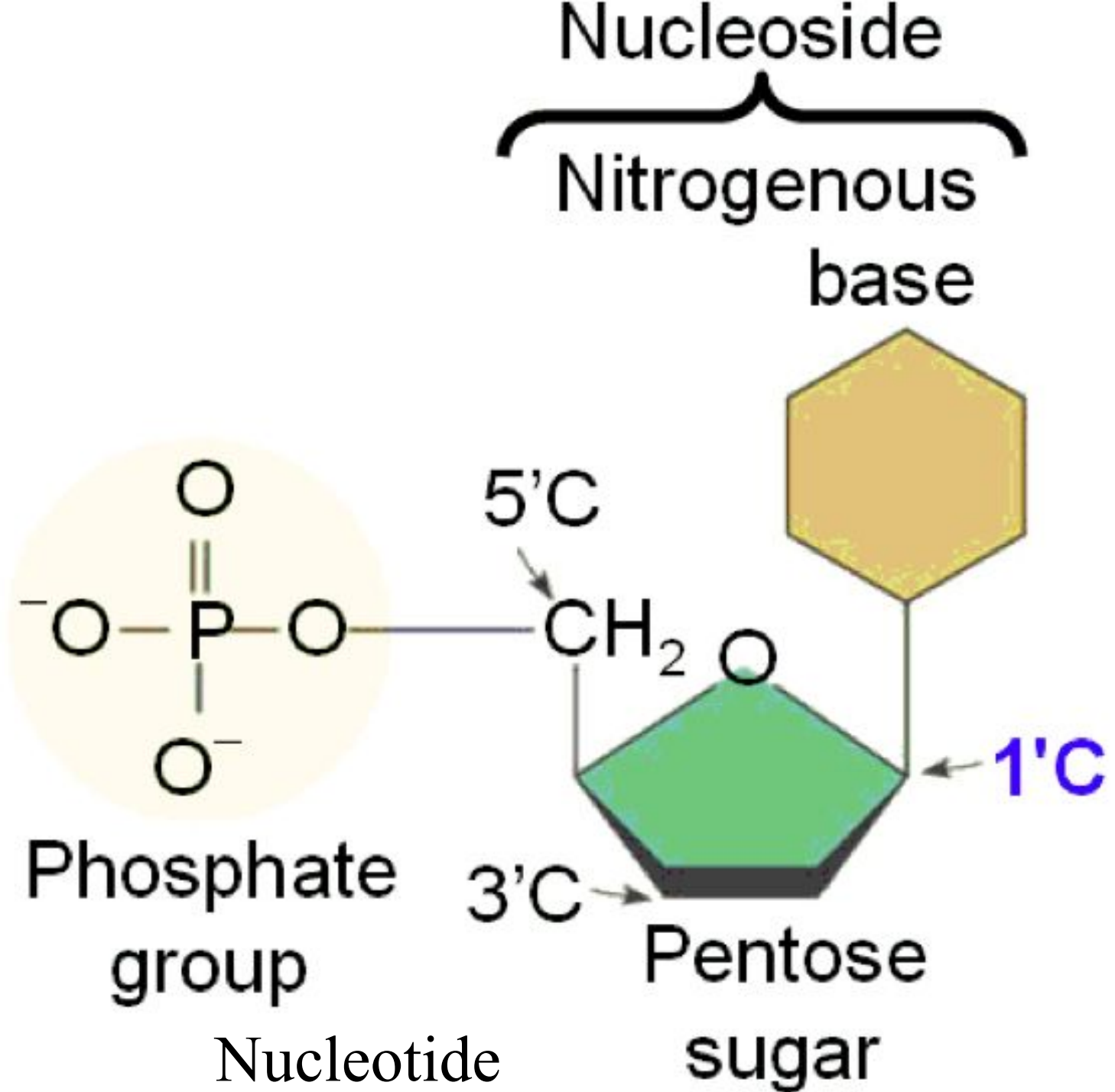


Franklin &
Gosling
5th March 1953
Type D

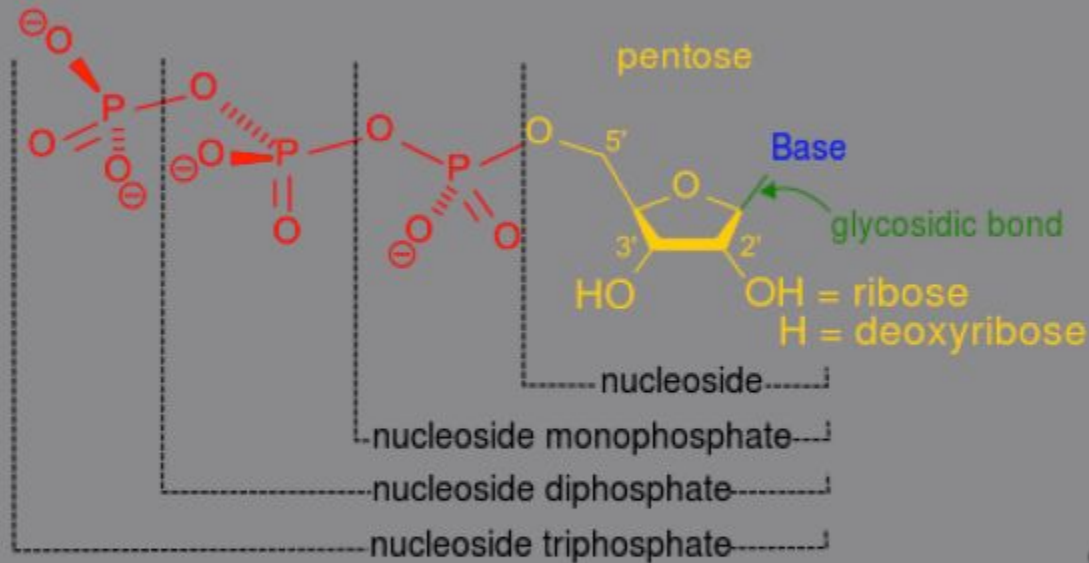
Plate 1

The structure of DNA

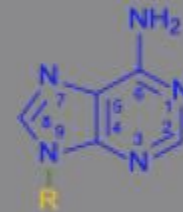




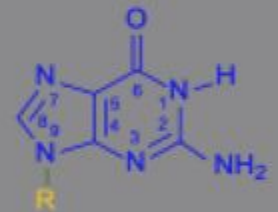
Structure of nucleic acids



Purines

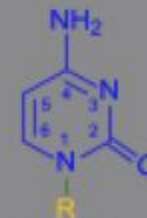


Adenine

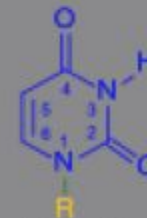


Guanine

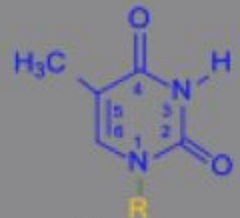
Pyrimidines



Cytosine



Uracil



Thymine

Chargaff's ratios

Table 11-1. Molar Properties of Bases* in DNAs from Various Sources

Organism	Tissue	Adenine	Thymine	Guanine	Cytosine
<i>Escherichia coli</i> (K12)	—	26.0	23.9	24.9	25.2
<i>Diplococcus pneumoniae</i>	—	29.8	31.6	20.5	18.0
<i>Mycobacterium tuberculosis</i>	—	15.1	14.6	34.9	35.4
Yeast	—	31.3	32.9	18.7	17.1
<i>Paracentrotus lividus</i> (sea urchin)	Sperm	32.8	32.1	17.7	18.4
Herring	Sperm	27.8	27.5	22.2	22.6
Rat	Bone marrow	28.6	28.4	21.4	21.5
Human	Thymus	30.9	29.4	19.9	19.8
Human	Liver	30.3	30.3	19.5	19.9
Human	Sperm	30.7	31.2	19.3	18.8

* Defined as moles of nitrogenous constituents per 100 g-atoms phosphate in hydrolysate.

SOURCE: E. Chargaff and J. Davidson, eds., *The Nucleic Acids*. Academic Press, 1955.

which are involved in ester linkages. This distortion of the phosphate group from the regular tetrahedral configuration is not supported by direct experimental evidence; unfortunately no precise structure determinations have been made of any phosphate di-esters. The distortion, which corresponds to a larger amount of double bond character for the inner oxygen atoms than for the oxygen atoms involved in the ester linkages, is a reason-

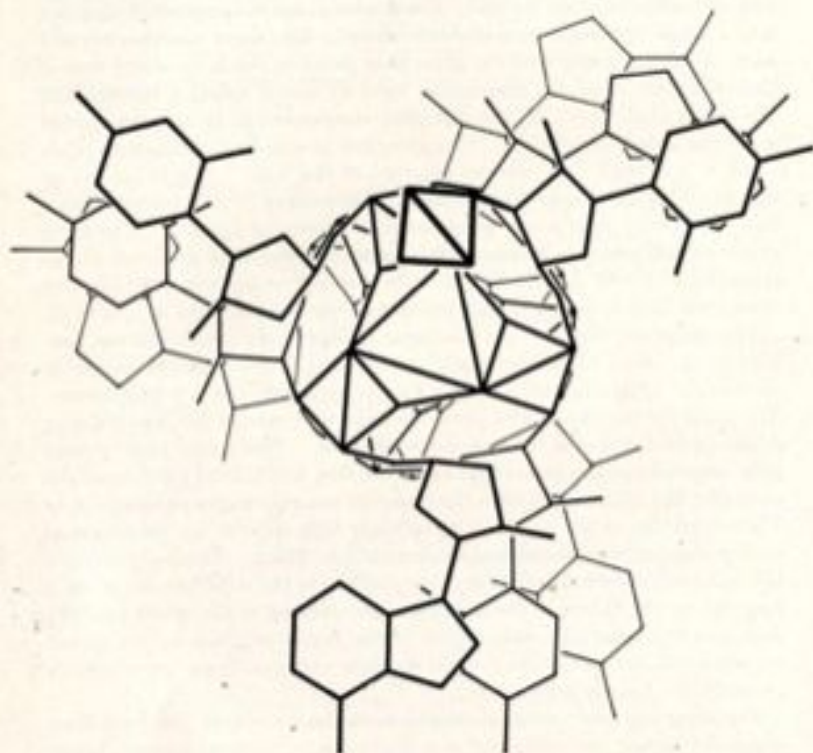


FIGURE 6

Plan of the nucleic acid structure, showing several nucleotide residues.

able one, and the assumed distances are those indicated by the observed values for somewhat similar substances, especially the ring compound S_8O_8 , in which each sulfur atom is surrounded by a tetrahedron of four oxygen atoms, two of which are shared with adjacent tetrahedra, and two unshared. The O—O distances within the phosphate tetrahedron are 2.32 Å (between the two inner oxygen atoms), 2.46 Å, 2.55 Å, and 2.60 Å. The

What makes this structure improbable?

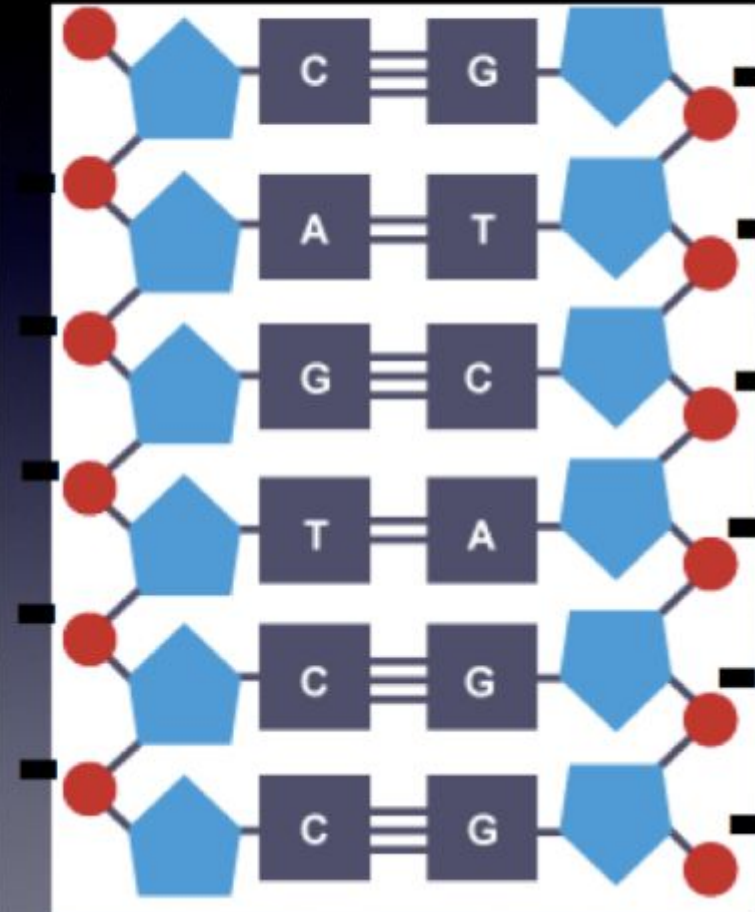
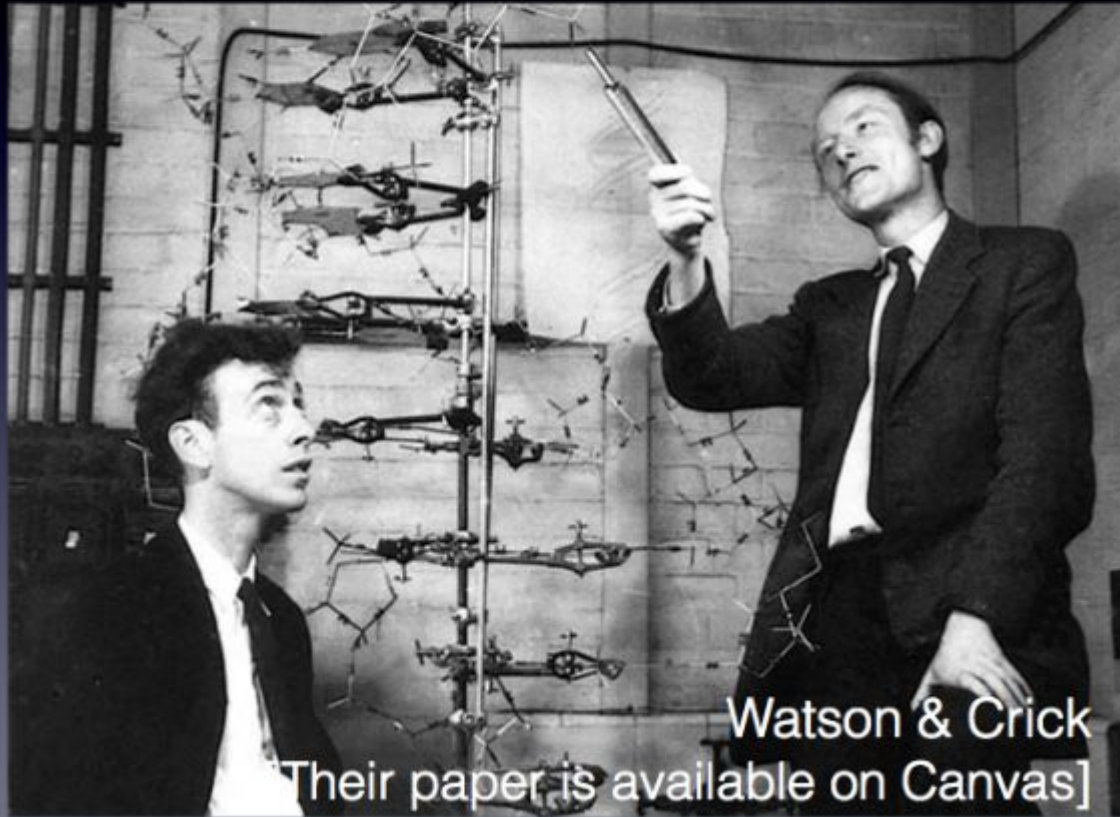
A.) outward-facing bases

B.) repelling electrostatic charge

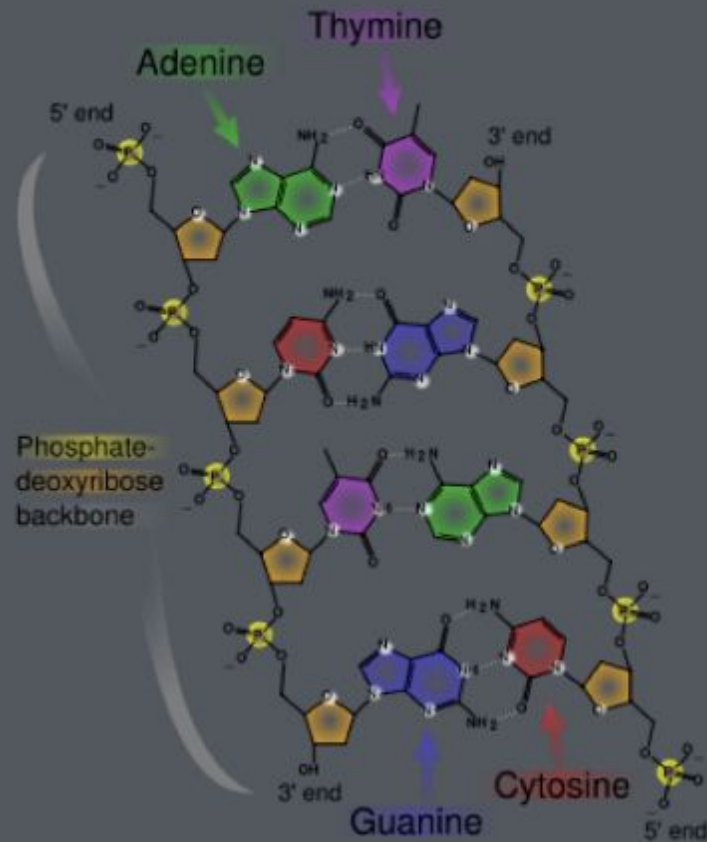
C.) steric clashes

Pauling & Corey, 1953

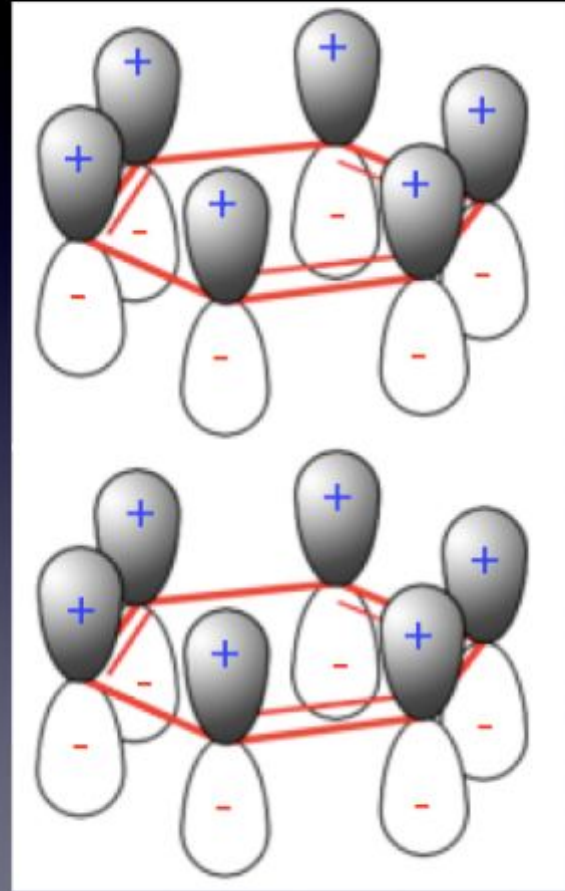
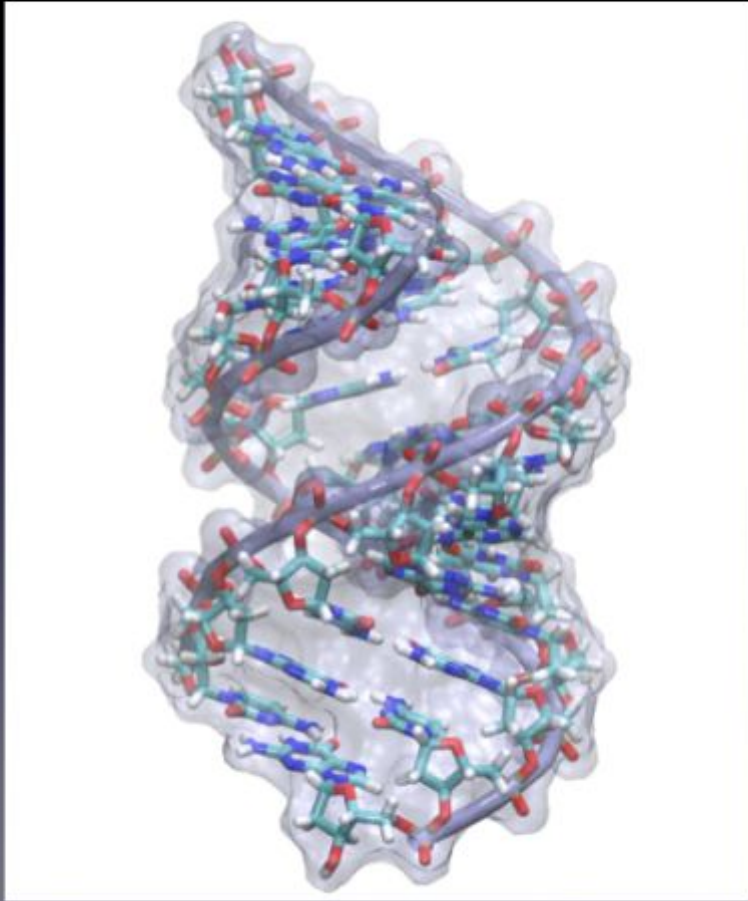
The double helix



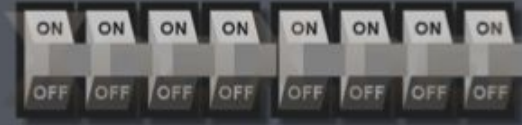
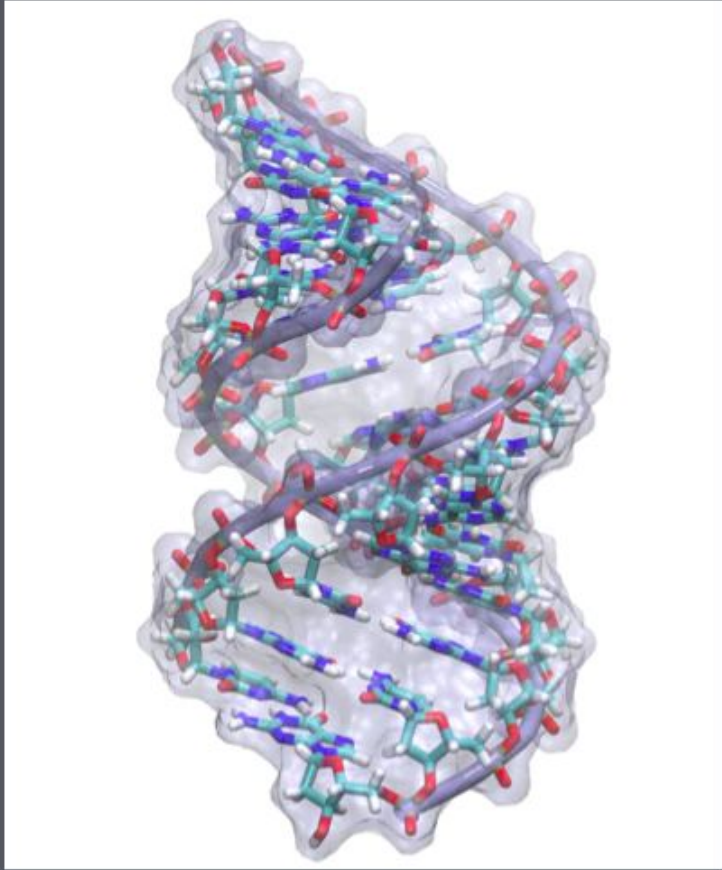
Stabilisation



Stabilisation II

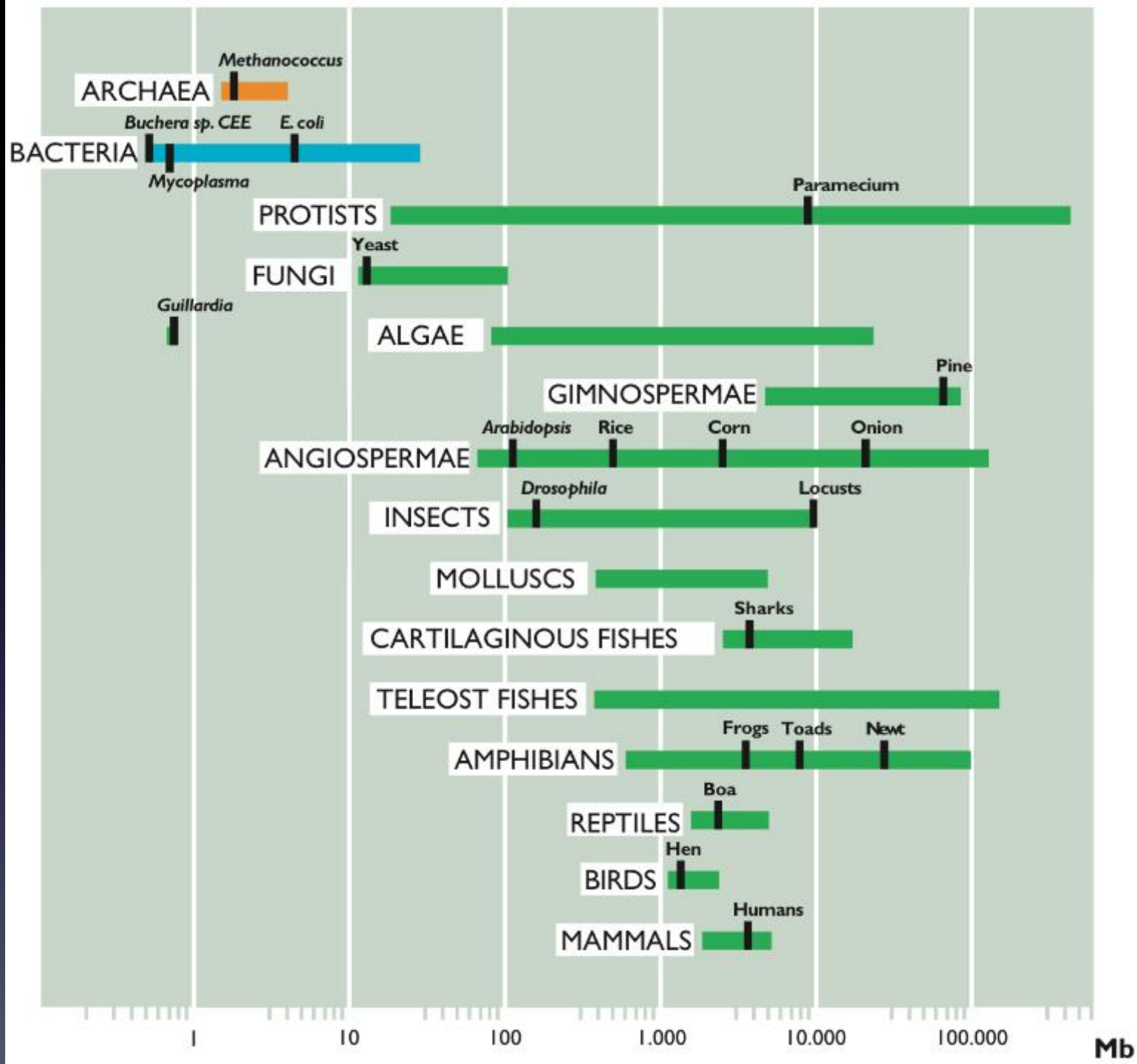


DNA function: Simplicity vs Complexity



$$2^8 = 256$$

$$4^8 = 65,536$$



An integrated encyclopedia of DNA elements in the human genome

The ENCODE Project Consortium*

The human genome encodes the blueprint of life, but the function of the vast majority of its nearly three billion bases is unknown. The Encyclopedia of DNA Elements (ENCODE) project has systematically mapped regions of transcription, transcription factor association, chromatin structure and histone modification. These data enabled us to assign biochemical functions for 80% of the genome, in particular outside of the well-studied protein-coding regions. Many discovered candidate regulatory elements are physically associated with one another and with expressed genes, providing new insights into the mechanisms of gene regulation. The newly identified elements also show a statistical correspondence to sequence variants linked to human disease, and can thereby guide interpretation of this variation. Overall, the project provides new insights into the organization and regulation of our genes and genome, and is an expansive resource of functional annotations for biomedical research.

The human genome sequence provides the underlying code for human biology. Despite intensive study, especially in identifying



ENCODE

Encyclopedia of DNA Elements

95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction (as assayed by bound ChIP-seq motifs or DNase I footprints),



8.2% of the Human Genome Is Constrained: Variation in Rates of Turnover across Functional Element Classes in the Human Lineage

Chris M. Rands¹, Stephen Meader¹, Chris P. Ponting^{1*}, Gerton Lunter^{2*}

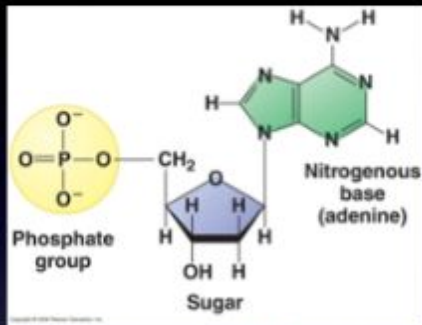
1 MRC Functional Genomics Unit, Department of Physiology, Anatomy, and Genetics, University of Oxford, Oxford, United Kingdom, **2** Wellcome Trust Centre for Human Genetics, University of Oxford, Oxford, United Kingdom

Abstract

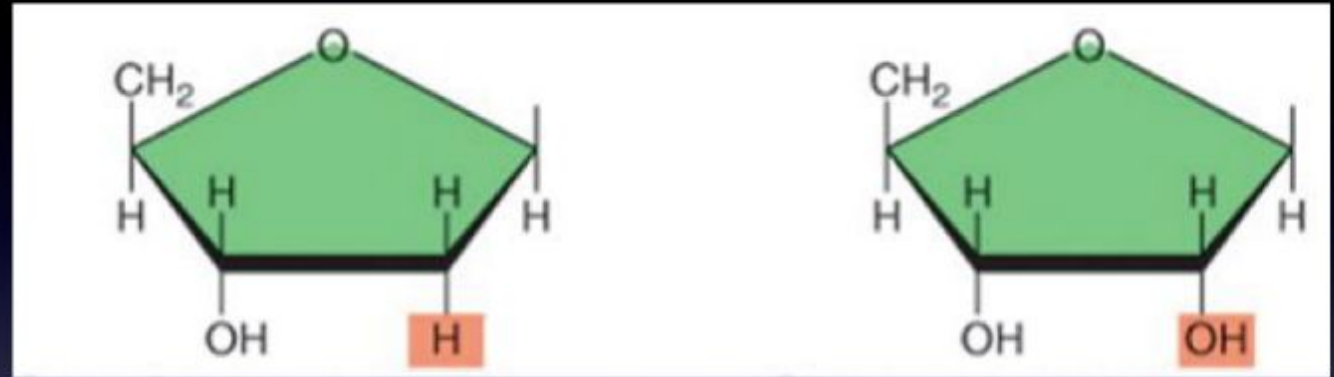
Ten years on from the finishing of the human reference genome sequence, it remains unclear what fraction of the human genome confers function, where this sequence resides, and how much is shared with other mammalian species. When addressing these questions, functional sequence has often been equated with pan-mammalian conserved sequence. However, functional elements that are short-lived, including those contributing to species-specific biology, will not leave a footprint of long-lasting negative selection. Here, we address these issues by identifying and characterising sequence that has been constrained with respect to insertions and deletions for pairs of eutherian genomes over a range of divergences. Within noncoding sequence, we find increasing amounts of mutually constrained sequence as species pairs become more closely related, indicating that noncoding constrained sequence turns over rapidly. We estimate that half of present-day noncoding constrained sequence has been gained or lost in approximately the last 130 million years (half-life in units of divergence time, $d_{1/2}=0.25-0.31$). While enriched with ENCODE biochemical annotations, much of the short-lived constrained sequences we identify are not detected by models optimized for wider pan-mammalian conservation. Constrained DNase 1 hypersensitivity sites, promoters and untranslated regions have been more evolutionarily stable than long noncoding RNA loci which have turned over especially rapidly. By contrast, protein coding sequence has been highly stable, with an estimated half-life of over a billion years ($d_{1/2}=2.1-5.0$). From extrapolations we estimate that 8.2% (7.1–9.2%) of the human genome is presently subject to negative selection and thus is likely to be functional, while only 2.2% has maintained constraint in both human and mouse since these species diverged. These results reveal that the evolutionary history of the human genome has been highly dynamic, particularly for its noncoding yet biologically functional fraction.

Citation: Rands CM, Meader S, Ponting CP, Lunter G (2014) 8.2% of the Human Genome Is Constrained: Variation in Rates of Turnover across Functional Element Classes in the Human Lineage. PLoS Genet 10(7): e1004525. doi:10.1371/journal.pgen.1004525

RNA = RiboNucleicAcid

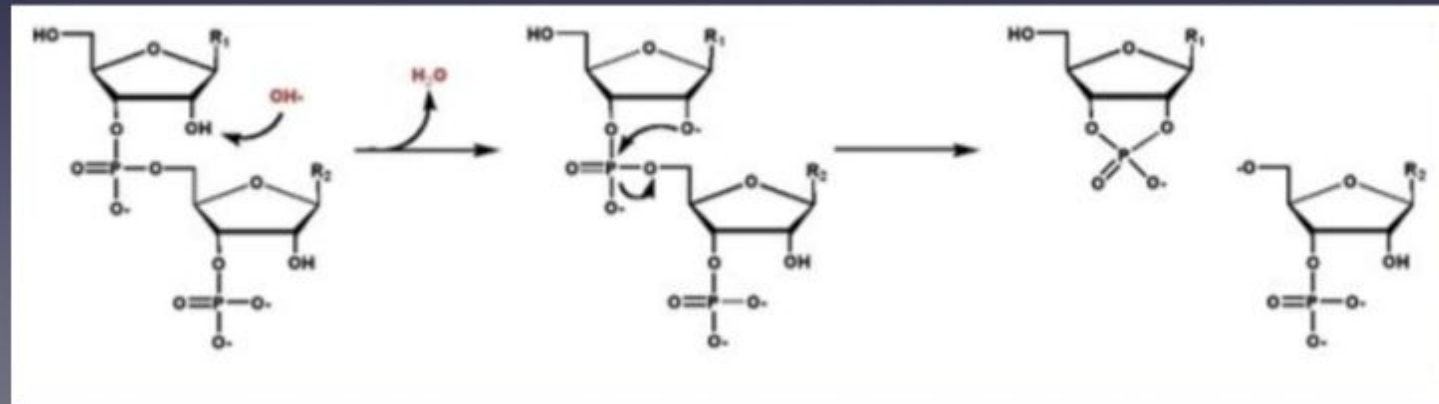


DNA nucleotide

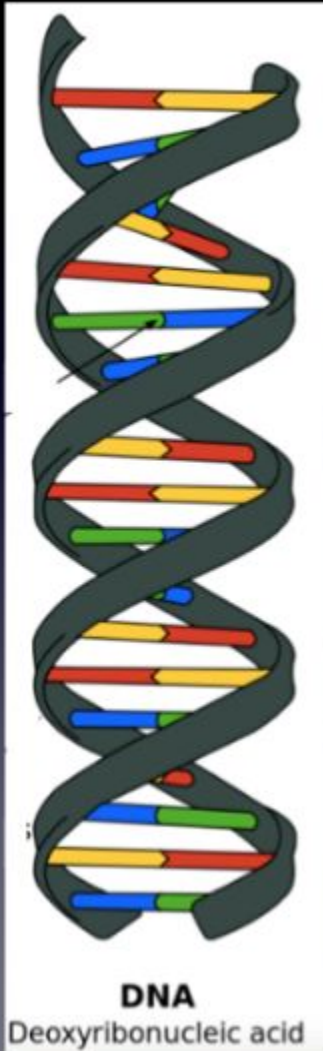


deoxyribose (in DNA)

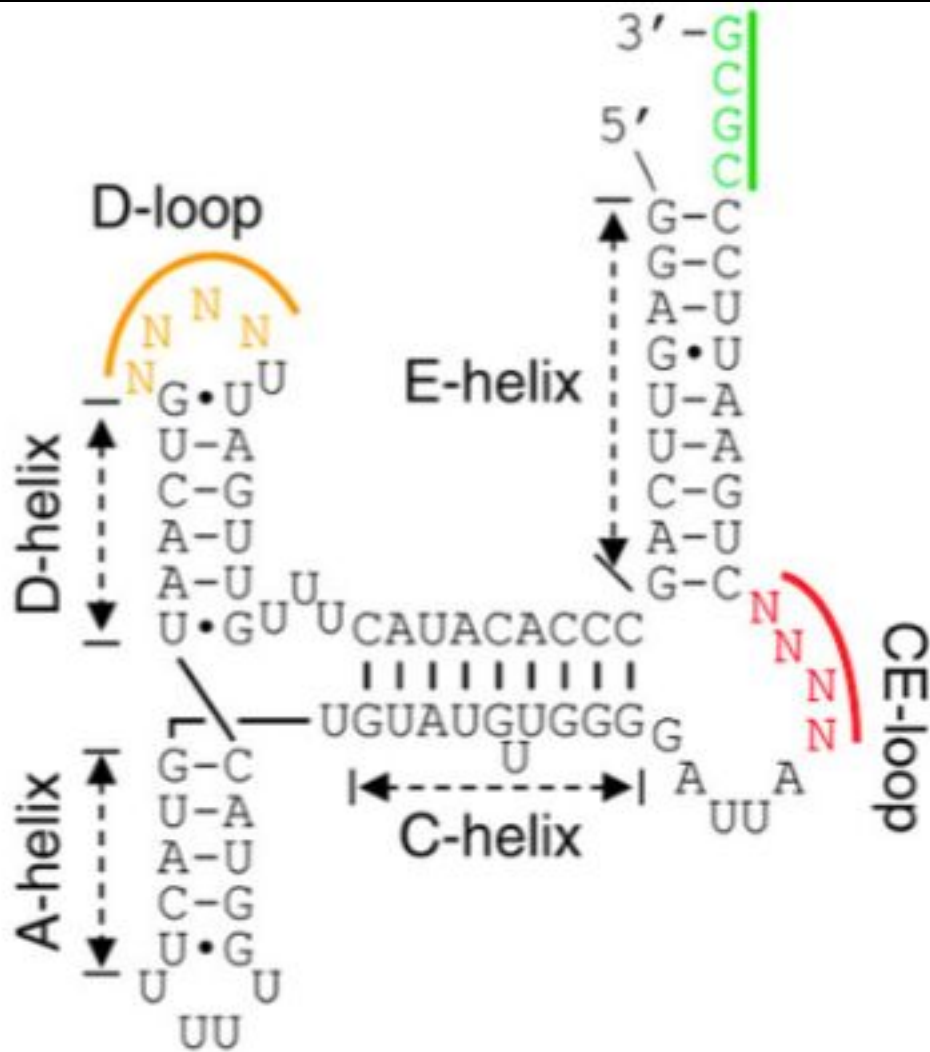
ribose (in RNA)



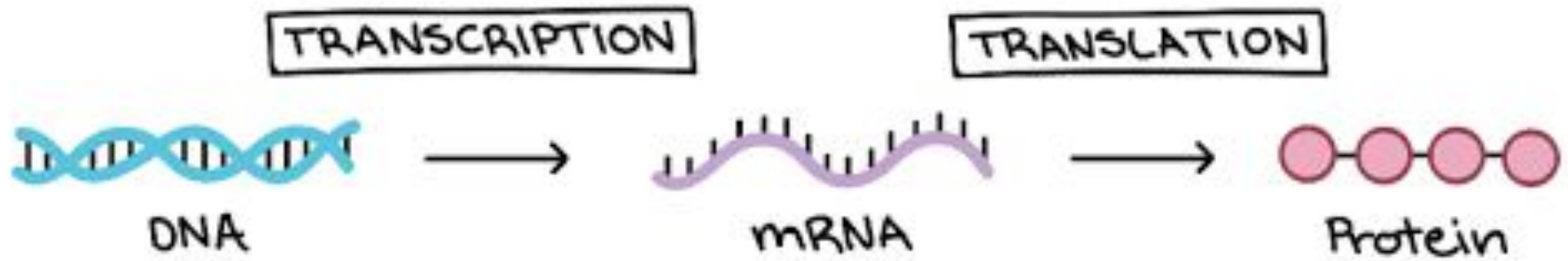
Structures



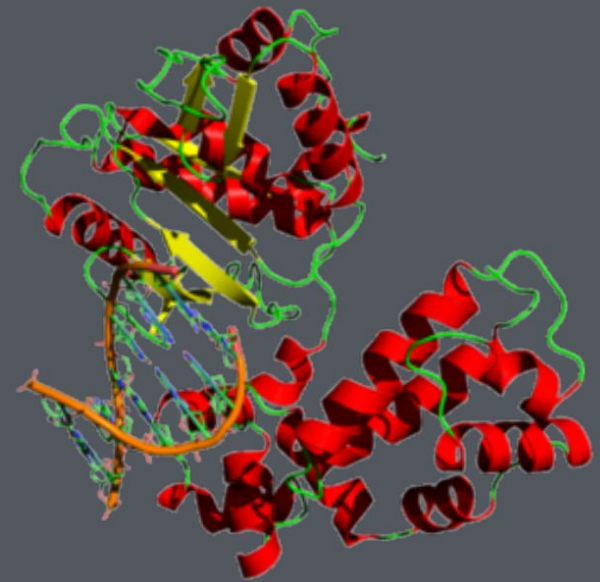
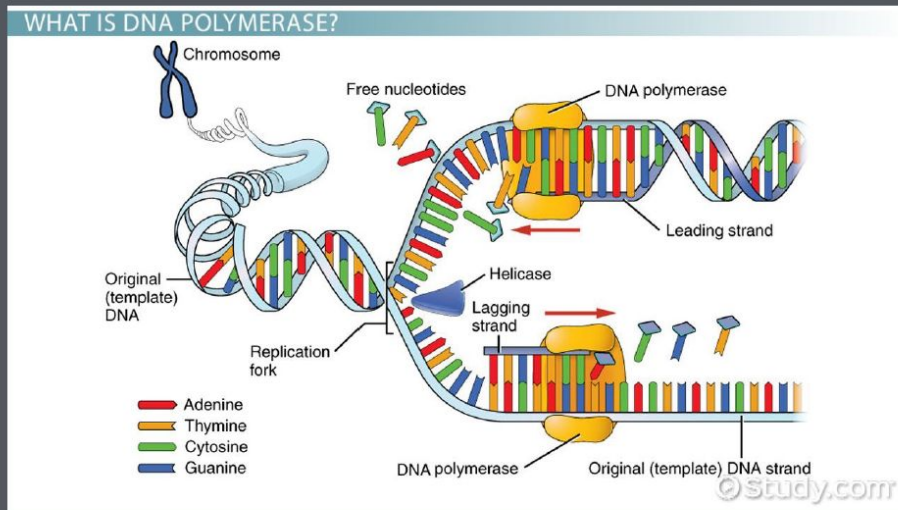
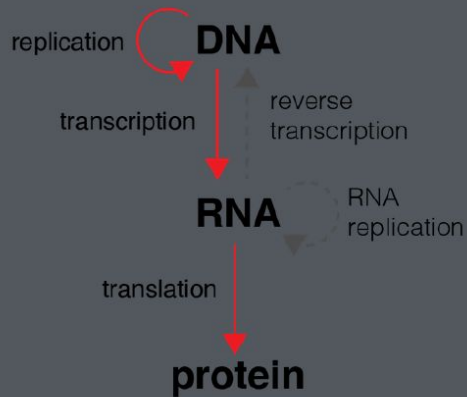
Self-Complimentarity



Central Dogma of Molecular Biology

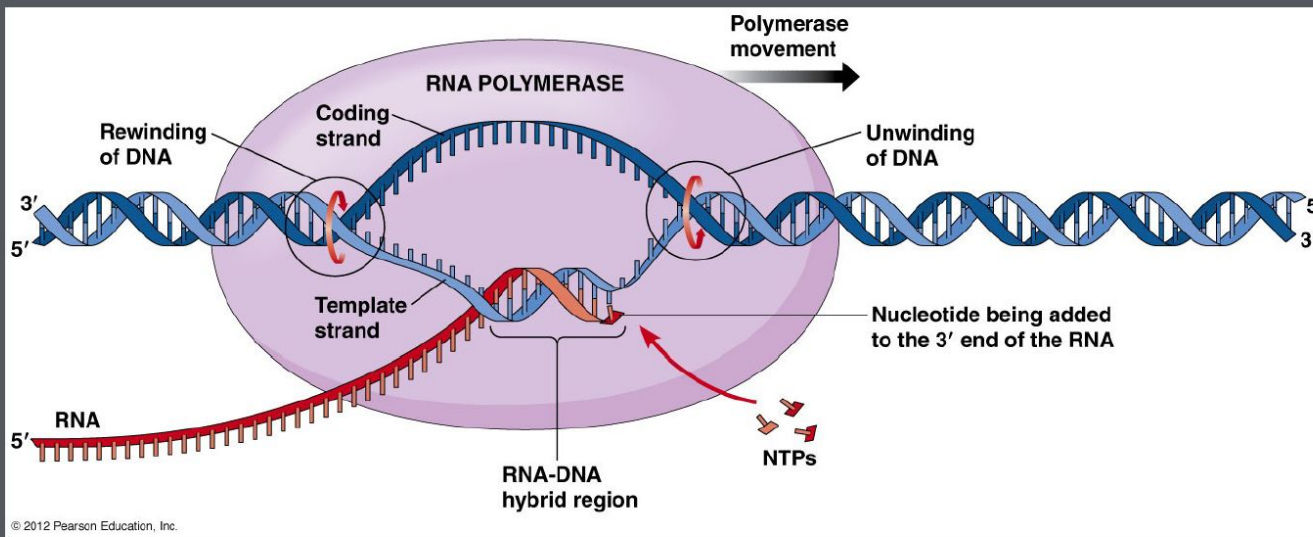
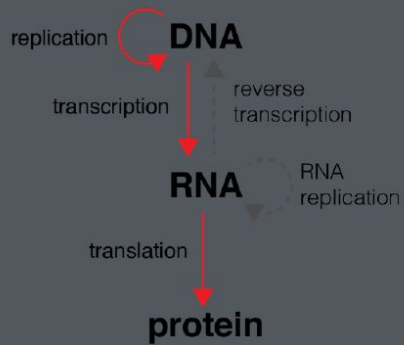


Replication

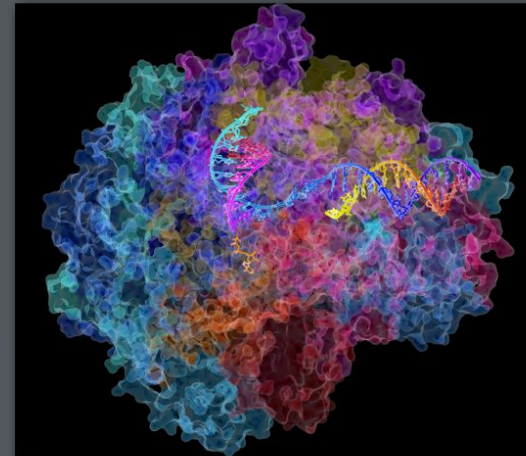


DNA Polymerase

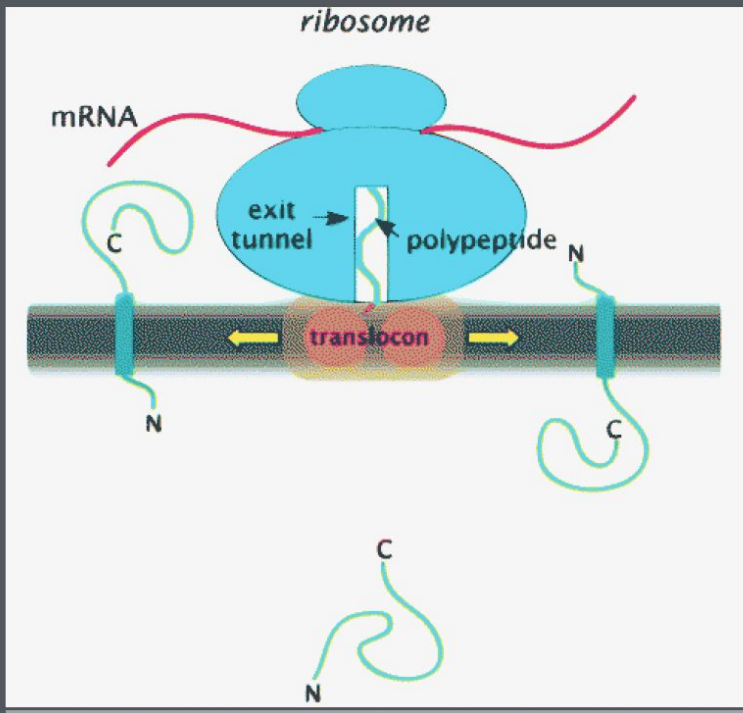
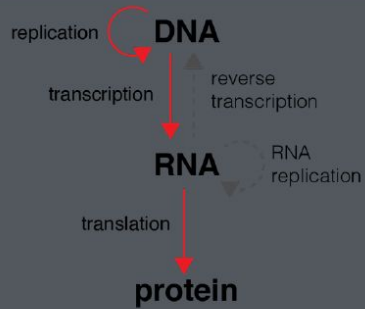
Transcription



Roger Kornberg 2006



Translation



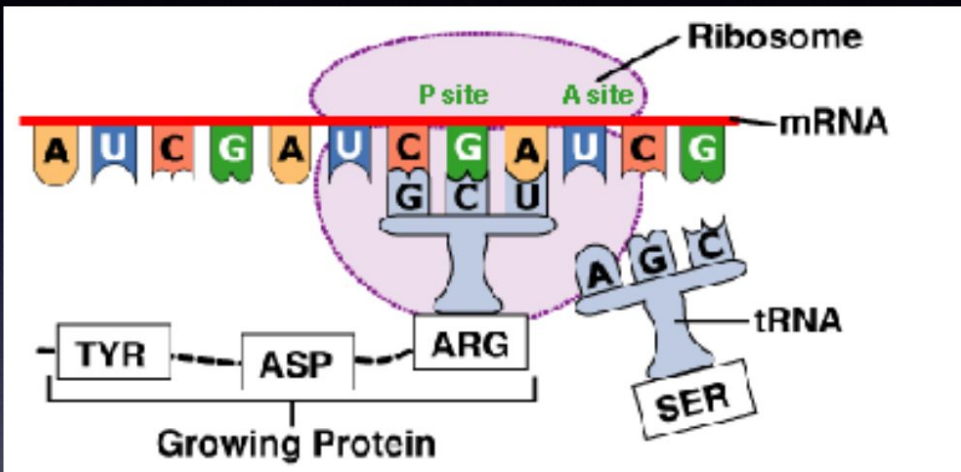
(White & von Heijne, 2005)

Ribosome



Tom Steitz, Peter Moore,
Venkatraman Ramakrishnan
Nobel 2009

The Genetic Code



		Second Letter				
		T	C	A	G	
First Letter	T	TTT } Phe TTC } TTA } Leu TTG }	TCT } Ser TCC } TCA } TCG }	TAT } Tyr TAC } TAA Stop TAG Stop	TGT } Cys TGC } TGA Stop TGG Trp	T C A G
	C	CTT } Leu CTC } CTA } CTG }	CCT } Pro CCC } CCA } CCG }	CAT } His CAC } CAA Gln CAG }	CGT } Arg CGC } CGA } CGG }	T C A G
	A	ATT } Ile ATC } ATA } ATG Met	ACT } Thr ACC } ACA } ACG }	AAT } Asn AAC } AAA Lys AAG }	AGT } Ser AGC } AGA Arg AGG }	T C A G
	G	GTT } Val GTC } GTA } GTG }	GCT } Ala GCC } GCA } GCG }	GAT } Asp GAC } GAA Glu GAG }	GGT } Gly GGC } GGA } GGG }	T C A G

Protein structure



Max Perutz & Hemoglobin

X-ray structure
Took 22 years...

